



CENTRALIZED LOGGING AND SECURITY MONITORING USING SPLUNK

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Centralized Logging and Security Monitoring Using Splunk

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Objective

This documentation describes the complete steps to forward **all Kali Linux systemd-journald logs** to **Splunk Enterprise installed on Rocky Linux** using **Splunk Universal Forwarder** and **rsyslog**.

Lab Environment

Component	Details
Splunk Server	Rocky Linux VM
Log Source	Kali Linux VM
Network Type	Host-Only Network
Log Transport	TCP 9997
Log Manager	systemd-journald + rsyslog
Forwarder	Splunk Universal Forwarder

Network Connectivity Verification

Check IP Address

On **Rocky Linux**:

ip add

```
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group de
fault qlen 1000
    link/ether 00:0c:29:cd:16:87 brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    inet 192.168.35.132/24 brd 192.168.35.255 scope global dynamic noprefixroute
ens192
        valid_lft 1278sec preferred_lft 1278sec
    inet6 fe80::20c:29ff:fe8d:1687/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

On **Kali Linux**:

ip add

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:01:66:40 brd ff:ff:ff:ff:ff:ff
    inet 192.168.35.131/24 brd 192.168.35.255 scope global dynamic noprefixroute eth1
        valid_lft 1106sec preferred_lft 1106sec
    inet6 fe80::cf42:642d:faa2:2cd6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Ensure both VMs are in the **same Host-Only subnet**.

Test Connectivity

From Kali:

ping <rocky-ip>

```
(kali㉿kali)-[~]
$ ping 192.168.35.132
PING 192.168.35.132 (192.168.35.132) 56(84) bytes of data.
64 bytes from 192.168.35.132: icmp_seq=1 ttl=64 time=2.02 ms
64 bytes from 192.168.35.132: icmp_seq=2 ttl=64 time=0.936 ms
64 bytes from 192.168.35.132: icmp_seq=3 ttl=64 time=1.09 ms
^C
— 192.168.35.132 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.936/1.349/2.020/0.478 ms
```

Expected result: Successful ICMP replies.

Splunk Server Configuration (Rocky Linux)

Start Splunk

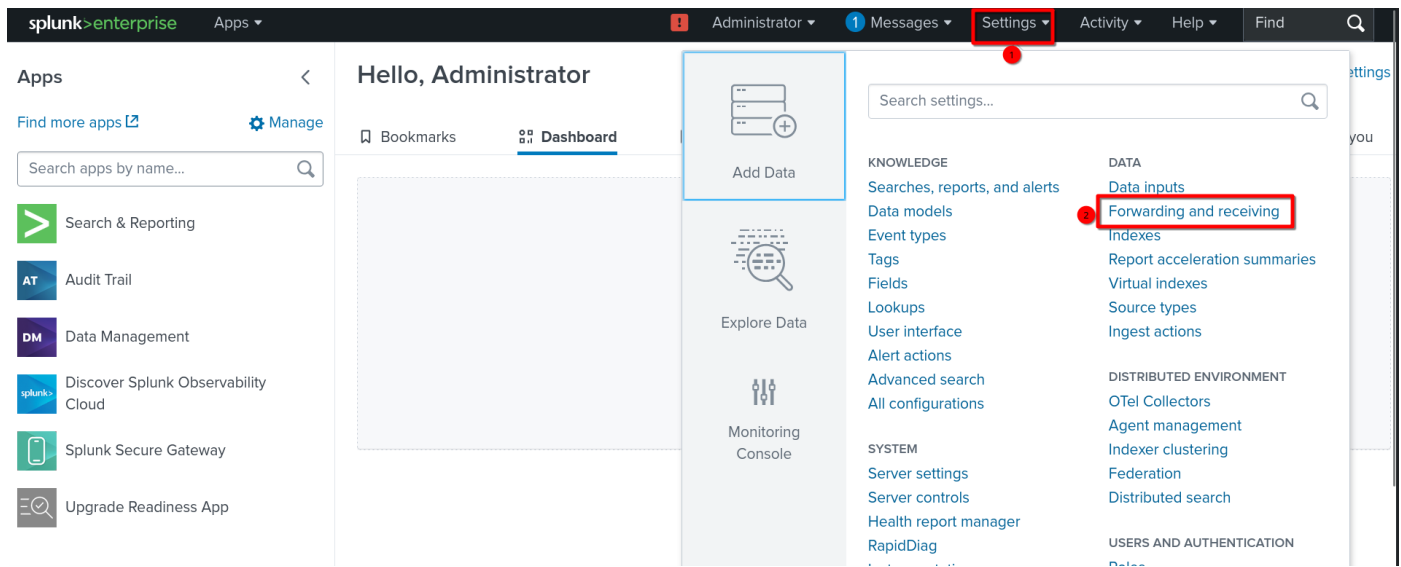
```
sudo /opt/splunk/bin/splunk start
```

```
[splunk@localhost ~]$ sudo /opt/splunk/bin/splunk start
[sudo] password for splunk:
The splunk daemon (splunkd) is already running.
```

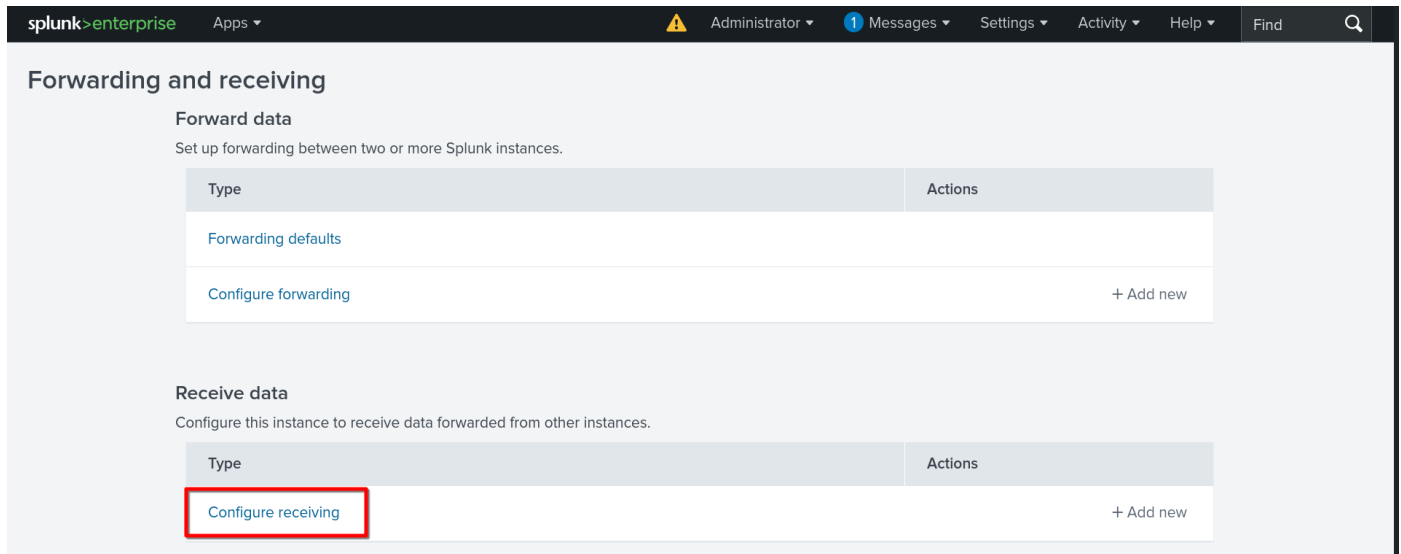
Enable Receiving Port

1. Go to Splunk GUI using browser:
<http://localhost:8000>
2. Type your credentials and login.

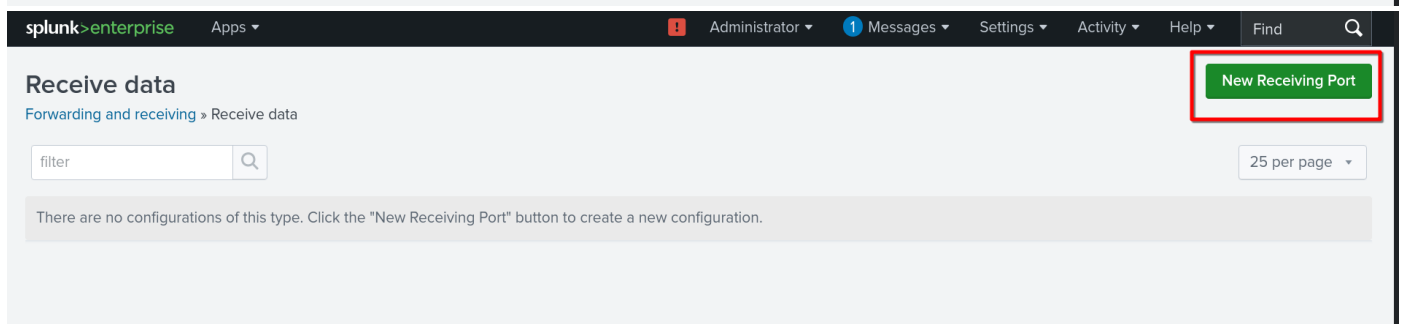




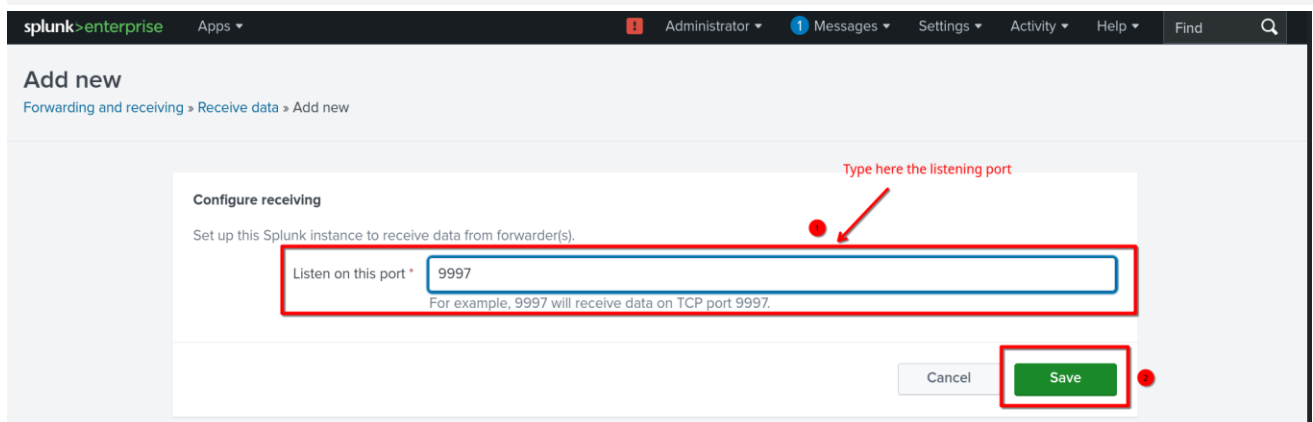
3.



4.

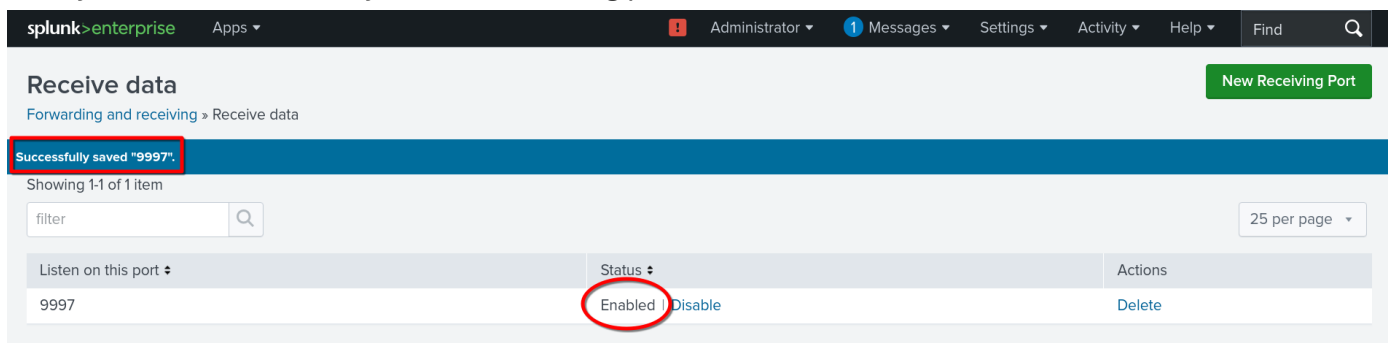


5.



6.

7. Now you have successfully added a listening port which is 9997



Configure Firewall

Allow the firewall to accept requests on the listening port.

```
sudo firewall-cmd --permanent --add-port=9997/tcp
sudo firewall-cmd --reload
```

```
[splunk@localhost ~]$ sudo firewall-cmd --permanent --add-port=9997/tcp
[sudo] password for splunk:
Warning: ALREADY_ENABLED: 9997:tcp
success
[splunk@localhost ~]$ sudo firewall-cmd --reload
success
```

Create Dedicated Index

Create an indexer for kali machine.

```
sudo /opt/splunk/bin/splunk add index kali
```

```
[splunk@localhost ~]$ sudo /opt/splunk/bin/splunk add index kali
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConfig]/cliVerifyServerName for details.
Index "kali" added.
```

Kali Linux Log Source Configuration

1. Enable Persistent journald Logging

```
sudo mkdir -p /var/log/journal
sudo systemctl restart systemd-journald
```

```
(kali㉿kali)-[~]
└─$ sudo mkdir -p /var/log/journal
[sudo] password for kali:

(kali㉿kali)-[~]
└─$ sudo systemctl restart systemd-journald

(kali㉿kali)-[~]
└─$
```

2. Install and Enable rsyslog

Here we will install Syslog service (demean) to collect logs from journald.

```
sudo apt update
```

```
sudo apt install rsyslog -y
```

```
(kali㉿kali)-[~]
$ sudo apt update
Hit:1 https://http.kali.org/kali kali-rolling InRelease
685 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali㉿kali)-[~]
$ sudo apt install rsyslog -y
rsyslog is already the newest version (8.2512.0-2).
The following packages were automatically installed and are no longer required:
  amass-common          libpgmep6t64          librav1e0.7          libwsutil16          python3-protobuf
  firmware-ti-connectivity libhdf4-0-alt         libsframe1           libx264-164          python3-pysmi
  gir1.2-girepository-2.0 libinstpatch-1.0-2    libsigsegv2          libx264-164          python3-wheel-whl
  libarmadillo14         libjs-jquery-ui       libsoup-2.4-1        linux-image-6.12.25-amd64 python3-xlrd
  libbluray2            libjs-underscore     libsoup2.4-common    python3-bluepy       python3-xlutils
  libbson-1.0-0t64       libmongoc-1.0-0t64   libsqlcipher1        python3-click-plugins python3-xlwt
  libdisplay-info2       libnet1              libtheora0           python3-gpg           python3-zombie-imp
  libgdal36             libobjc-14-dev        libtheoraec1         python3-kismetcapturebtgeiger samba-ad-dc
  libgdata-common        libogdi4.1           libtheoraenc1        python3-kismetcapturefreaklabszigbee samba-ad-provision
  libgdata22            libplacebo349         libudfread0          python3-kismetcapturertl433  samba-dsdb-modules
  libgeos3.13.1         libportmidi0         libvpx9              python3-kismetcapturertladsb
  libgirepository-1.0-1 libqt5ct-common1.8    libwireshark18       python3-kismetcaptureertlamr
  libpgme11t64          libradare2-5.0.0t64  libwiretap15         python3-packaging-whl
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 685

(kali㉿kali)-[~]
$
```

3. Enable Syslog Service

```
sudo systemctl enable rsyslog
```

```
sudo systemctl start rsyslog
```

```
(kali㉿kali)-[~]
$ sudo systemctl enable rsyslog

(kali㉿kali)-[~]
$ sudo systemctl start rsyslog

(kali㉿kali)-[~]
$
```


4. Verify

systemctl status rsyslog

```
(kali@kali)-[~]
$ systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-01-10 14:08:40 EST; 42min ago
     Invocation: e14d3f0d3a02401fbae4915f92e717ed
   TriggeredBy: ● syslog.socket
      Docs: man:rsyslogd(8)
            man:rsyslog.conf(5)
            https://www.rsyslog.com/doc/
    Main PID: 2094 (rsyslogd)
       Tasks: 4 (limit: 2107)
      Memory: 1.2M (peak: 2.6M)
         CPU: 71ms
      CGroup: /system.slice/rsyslog.service
             └─2094 /usr/sbin/rsyslogd -n -iNONE

Jan 10 14:08:40 kali systemd[1]: Starting rsyslog.service - System Logging Service...
Jan 10 14:08:40 kali rsyslogd[2094]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2512.0]
Jan 10 14:08:40 kali systemd[1]: Started rsyslog.service - System Logging Service.
Jan 10 14:08:40 kali rsyslogd[2094]: [origin software="rsyslogd" swVersion="8.2512.0" x-pid="2094" x-info="https://www.rsyslog.com"] start
```

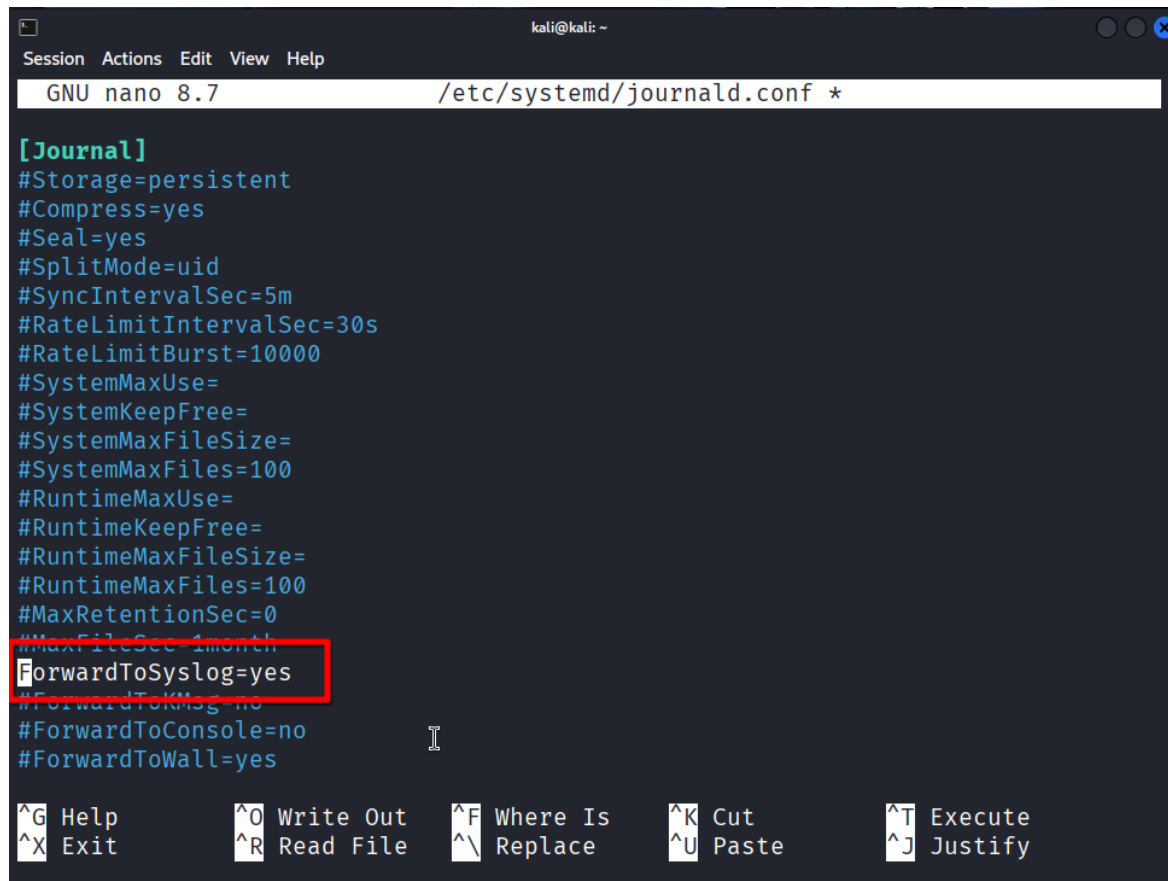
5. Forward journald Logs to rsyslog

5.1 Edit configuration file

sudo nano /etc/systemd/journald.conf

5.2 Set

ForwardToSyslog=yes



```
kali@kali: ~
Session Actions Edit View Help
GNU nano 8.7 /etc/systemd/journald.conf *

[Journal]
#Storage=persistent
#Compress=yes
#Seal=yes
#SplitMode=uid
#SyncIntervalSec=5m
#RateLimitIntervalSec=30s
#RateLimitBurst=10000
#SystemMaxUse=
#SystemKeepFree=
#SystemMaxFileSize=
#SystemMaxFiles=100
#RuntimeMaxUse=
#RuntimeKeepFree=
#RuntimeMaxFileSize=
#RuntimeMaxFiles=100
#MaxRetentionSec=0
#MaxFileSec=1month
ForwardToSyslog=yes
#ForwardToKmsg=no
#ForwardToConsole=no
#ForwardToWall=yes

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify
```

- Press CTRL+O then ENTER to Save.
- Press CTRL+X to exit.

5.3 Restart Services

```
sudo systemctl restart systemd-journald
```

```
sudo systemctl restart rsyslog
```

```
(kali㉿kali)-[~]  
$ sudo systemctl restart systemd-journald  
  
(kali㉿kali)-[~]  
$ sudo systemctl restart rsyslog
```

Splunk Universal Forwarder Configuration (Kali Linux)

Download UF

```
wget -O splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.tgz  
"https://download.splunk.com/products/universalforwarder/releases/10.0  
.2/linux/splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.tgz"
```

Extract & Install

```
sudo tar -xvzf splunkforwarder.tgz -C /opt
```

Start Forwarder

```
cd /opt/splunkforwarder/bin  
sudo ./splunk start --accept-license
```

```
(kali㉿kali)-[~]  
$ cd /opt/splunkforwarder/bin  
  
(kali㉿kali)-[/opt/splunkforwarder/bin]  
$ sudo ./splunk start --accept-license  
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-259.s  
o)  
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-259.s  
o)  
  
Splunk> See your world. Maybe wish you hadn't.  
  
Checking prerequisites ...  
  Checking mgmt port [8089]: open  
New certs have been generated in '/opt/splunkforwarder/etc/auth'.  
  Checking conf files for problems ...  
    Done  
  Checking default conf files for edits ...  
Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64-manifest'  
  All installed files intact.  
    Done  
All preliminary checks passed.  
  
Starting splunk server daemon (splunkd) ...  
Done
```

It might ask you for credentials, add yours.

Configure Splunk Receiver

```
sudo /opt/splunkforwarder/bin/splunk add forward-server <rocky-  
ip>:9997
```

```
(kali㉿kali)-[/opt/splunkforwarder/bin]  
$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.35.132:9997  
Your session is invalid. Please login.  
Splunk username: kali  
Password: ← Type here your credentials  
Added forwarding to: 192.168.35.132:9997.
```

Add syslog as Input

```
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog -index kali
sudo /opt/splunkforwarder/bin/splunk restart
```

```
(kali@kali)-[/opt/splunkforwarder/bin]
$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog -index kali
Added monitor of '/var/log/syslog'.

(kali@kali)-[/opt/splunkforwarder/bin]
$ sudo /opt/splunkforwarder/bin/splunk restart
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-259.so)
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-259.so)
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
Stopping splunk helpers ...

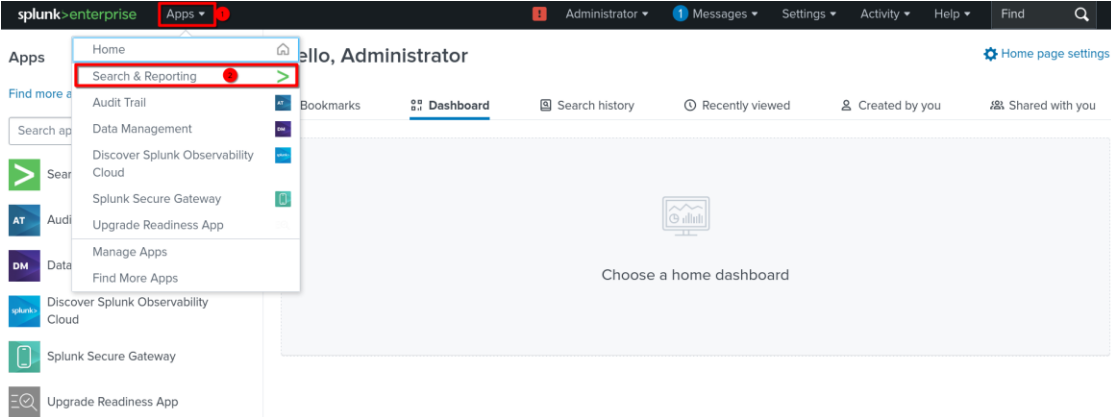
Done.
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-259.so)

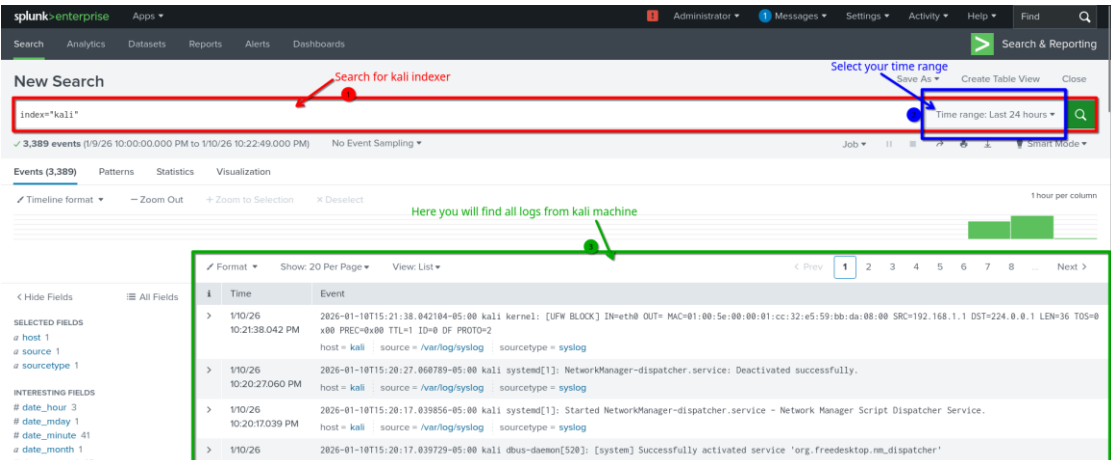
Splunk> See your world. Maybe wish you hadn't.

Checking prerequisites...
Checking mgmt port [8089]: open
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done
```

Verification on Splunk Server

1. 

2. 

The search results show logs from the kali machine, including:

- 2026-01-10T15:21:38.042 PM kali kernel: [UFW BLOCK] IN=eth0 OUT= MAC=81:00:5e:00:00:01 CC=32:e5:59:bb:da:88:00 SRC=192.168.1.1 DST=224.0.0.1 LEN=36 TOS=0
- 2026-01-10T15:20:27.060 PM kali system[1]: NetworkManager-dispatcher.service: Deactivated successfully.
- 2026-01-10T15:20:17.039 PM kali system[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
- 2026-01-10T15:20:17.039 PM kali dbus-daemon[520]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'